

Understanding Stress Factors Among University Students

*A data-driven exploration of how stress affects
academic performance and student well-being*

Madiha Safi · 2 Datasets · 250 Students · pandas · Plotly · DuckDB · scikit-learn

Why does this matter?

35%

of students
report depression

34%

of students
report anxiety

33%

experience
panic attacks

Only 6%

seek professional
treatment

University students face mounting pressure from academic workloads, social expectations, and lifestyle demands. Yet universities lack clear, data-backed insights into what drives stress — so interventions remain generic and ineffective.

Who is affected:

University Students

Academic Institutions

Mental Health Services

Policymakers

Two datasets · 250 students · 21 variables

Dataset 1 — Stress Factors & Mental Health

100 students · 16 variables

- ▶ Sleep Quality (1–5 Likert)
- ▶ Study Load (1–5 Likert)
- ▶ Headache Frequency
- ▶ Academic Performance
- ▶ Stress Level
- ▶ Age
- ▶ Course
- ▶ Year of Study
- ▶ CGPA
- ▶ Marital Status
- ▶ Gender
- ▶ Extracurricular Activities
- ▶ Depression
- ▶ Anxiety
- ▶ Panic Attack
- ▶ Sought Treatment

Dataset 2 — Stress Level Dataset

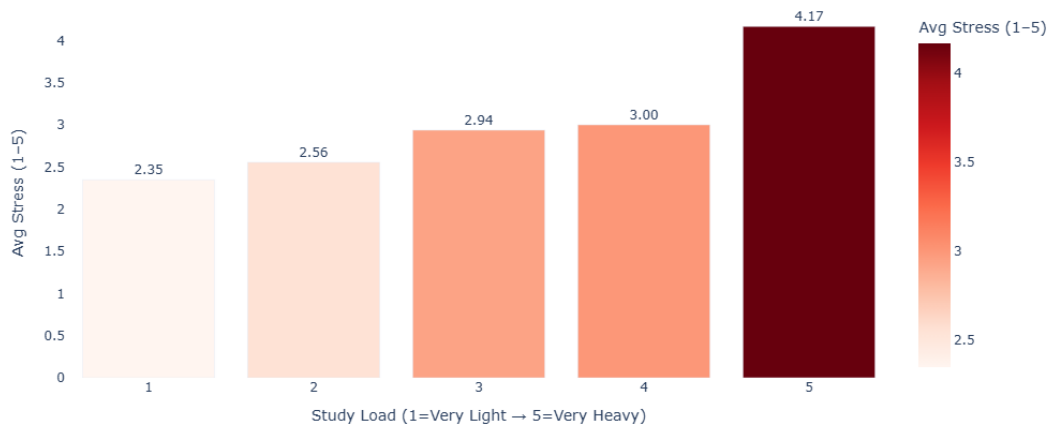
150 students · 12 variables

- ▶ Depression
- ▶ Sleep Quality
- ▶ Academic Performance
- ▶ Study Load
- ▶ Extracurricular Activities
- ▶ Stress Level
- ▶ Anxiety
- ▶ Living Conditions
- ▶ Future Career Concerns
- ▶ Social Support
- ▶ Peer Pressure
- ▶ Teacher–Student Relationship

◆ Shared across both datasets: Sleep Quality · Study Load · Academic Performance · Extracurricular Activities · Stress Level

What drives stress? Study Load is the #1 predictor

Study Load vs Average Stress Level (Dataset 1)



Correlation with Stress Level

Study Load (DS1)

+0.36 Strongest predictor

Depression (encoded) (DS1)

+0.59 With depression

Future Career Concerns (DS2)

+0.74 Top DS2 predictor

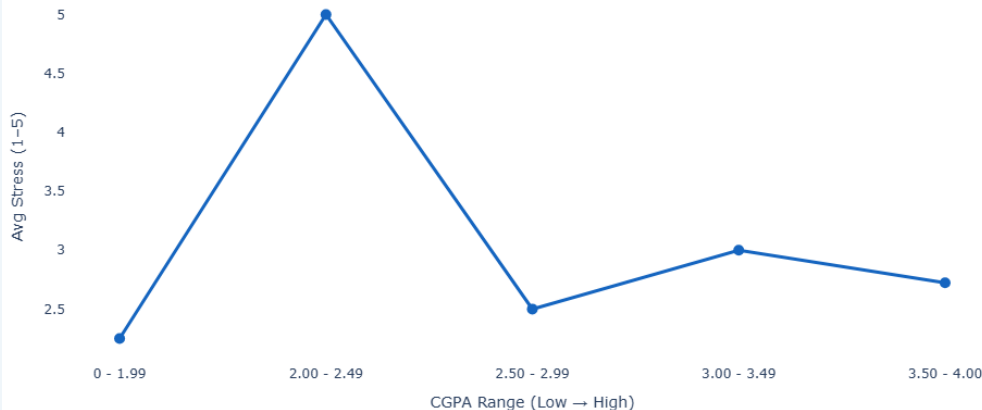
Sleep Quality (DS1)

+0.15 Positive but weaker

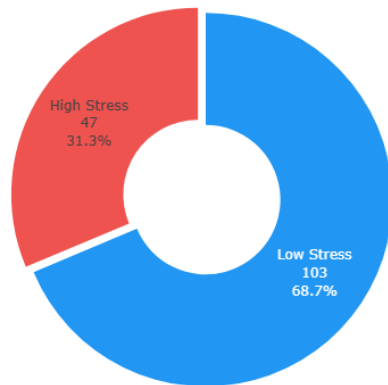
Key Finding 1: Heavy study load (5/5) students report 77% higher average stress than light-load (1/5) students (4.17 vs 2.35). Study load has the strongest lifestyle correlation with stress.

Mental health & career anxiety amplify stress significantly

CGPA vs Average Stress Level



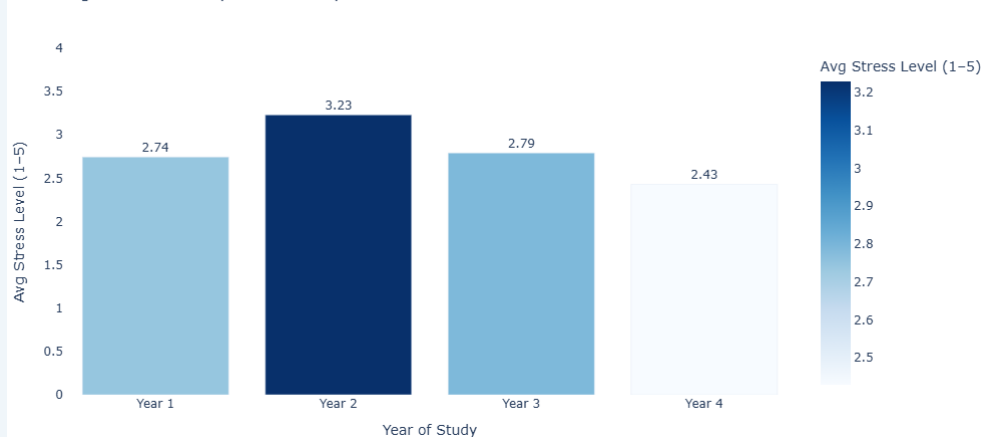
Proportion of Low vs High Stress — Dataset 2 (n=150)



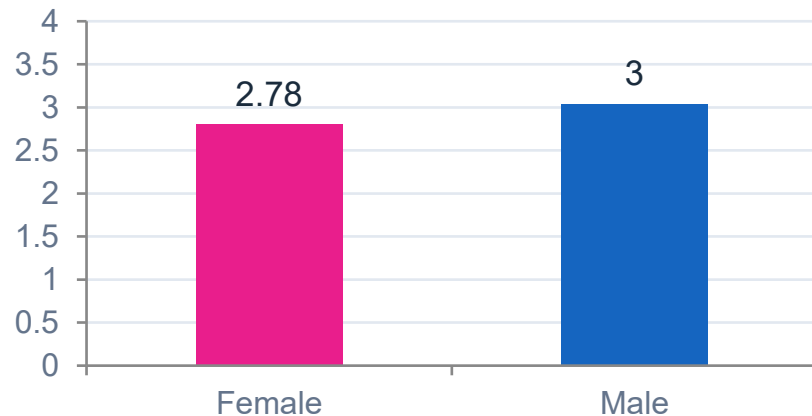
Key Finding 2: The lower the CGPA the higher the Stress level.

Who is most at risk? Year 2 students & male students show highest stress

Average Stress Level by Year of Study



Avg Stress by Gender



DuckDB Query Finding:

Year 2 students

3.23 avg stress

Highest of all years — new academic demands hit

Year 4 students

2.43 avg stress

Lowest — adapted to academic life

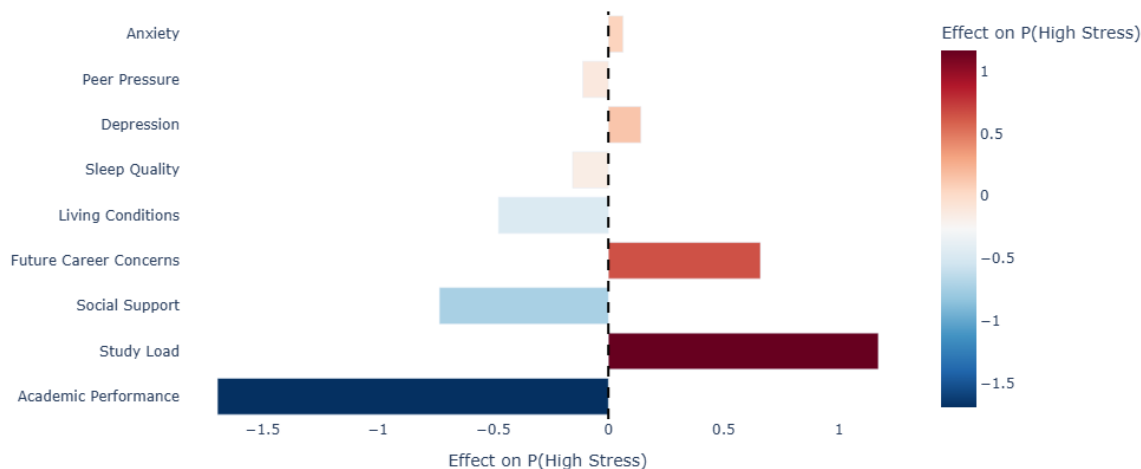
11 students

High stress + depression

Identified via SQL filtering (stress ≥ 4 AND depression = Yes)

Regression confirms: what predicts high stress most?

Logistic Regression — Predicting High vs Low Stress (Accuracy = 96.7%)



Tools & Methods

Python + pandas

Data loading & cleaning

Plotly

14 interactive visualizations

scikit-learn

Linear & logistic regression

DuckDB

SQL queries on DataFrames

Model Accuracy: 96.7% · R^2 (linear): 19.6%

The One-Sentence Recommendation

"Because study load and mental health pressure are the strongest predictors of high stress, we recommend universities introduce structured workload monitoring, mandatory mental health check-ins, and career guidance programmes targeting Year 1–3 students."



University Wellness Offices

- ▶ Mandatory mental health check-ins for Year 1 & Year 3 students
- ▶ Early identification: flag students with stress ≥ 4 AND depression
- ▶ Increase counsellor visibility
— only 6% currently seek help



Academic Schedulers

- ▶ Reduce back-to-back high-load modules in Years 2–3
- ▶ Cap weekly study load at sustainable levels
- ▶ Space high-stakes deadlines across the semester



Student Affairs / Policy

- ▶ Career counselling programmes for future anxiety (+0.74 correlation)
- ▶ Social support programmes (−0.69 correlation with stress)
- ▶ Data-driven intervention budgets based on this analysis

What changes if we act on this?

Potential Impact

Targeted support for Year 2 students
reduces peak-stress semester dropout

Career guidance programmes reduce
+0.74 anxiety correlation with stress

Earlier mental health intervention,
from 6% treated to meaningful numbers

Universities gain evidence base for
resource allocation decisions

Limitations & Next Steps

⚠ 100–150 students; single university
→ Scale to multi-university dataset

⚠ Self-reported data, response bias
→ Add objective measures (GPA, attendance)

⚠ Snapshot study, no time series
→ Track same students across a semester

⚠ Binary stress scale in DS2 limits precision
→ Build Streamlit dashboard for counsellors

250

students analysed

21

variables studied

96.7%

model accuracy

2.4×

higher depression
in high-stress group

The Answer:

"Because study load & mental health pressure are the strongest predictors of student stress, universities should act now, starting with Year 2 students."

Tools: Python · pandas · Plotly · scikit-learn · DuckDB · Google Colab · GitHub

Madiha Safi · Practical Data Science: Tools for Social Change · Final Project

Thank You